

Pooja Ravi

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EDUCATION

MSc, Applied Computing

2025

University of Toronto, Canada

Awards: Vector Scholarship in Artificial Intelligence (2023-24) | Mitacs Globalink Graduate Fellowship.

BTech, Computer Science and Engineering

2023

SRM Institute of Science and Technology, India

RESEARCH EXPERIENCE

Researcher | University of Toronto (TCIG)

Apr 2026 - Present

Supervised by David B. Lindell

In-hand robotic object manipulation

- **Action recovery.** Ray-traced query points on hand links, palm, and wrist; tracked them with CoTracker; then optimized framewise joint angles to align reprojected points with the tracks, warm-starting from previous-frame angles until convergence.
- **Physics replay.** Executed recovered actions in MuJoCo with contact and gravity; passed a frame from each replay back as the next video-generation cycle's initial frame, closing the loop.
- **Closed-loop pipeline.** Built CoTracker-based hand tracking and inverse-kinematics solve stages for ORCA Hand v1, then integrated them with a fine-tuned object-motion-conditioned video model to connect generated video to joint-space actions.

ProxyPose

- **Benchmarking.** Engineered a unified 6-DoF benchmark across HO3D, YCBinEOAT, and YCB-Video; standardized meshes, RGB/depth frames, pose annotations, and metadata in a consistent evaluation format.
- **Evaluation.** Implemented task-aligned metrics and dataset-specific evaluation pipelines; integrated competing methods, ran cross-dataset experiments, and analyzed comparative results.

PUBLICATIONS

1. R. Zhang, F. Taubner, **P. Ravi**, K. N. Kutulakos, and D. B. Lindell. **ProxyPose: 6-DoF Pose Tracking via Video-to-Video Translation**. NeurIPS 2026; arXiv:2607.06555.
2. **P. Ravi**, S. Roy, I. Dutta, and K. Kottursamy. **Attention Mechanism, Linked Networks, and Pyramid Pooling Enabled 3D Biomedical Image Segmentation**. IEEE/ACIS SNPD, 2022.
3. **P. Ravi** and A. Dewan. **Real-time Multi-Module Student Engagement Detection System**. Proceedings of ICCIS 2022, Lecture Notes in Networks and Systems 686, 2022.

APPLIED MACHINE LEARNING EXPERIENCE

Machine Learning Engineer | Q2 Software

May 2024 - Mar 2026

- Led cheque-signature verification across 200K images, training variational autoencoder embeddings and Siamese networks with triplet and contrastive losses; fine-tuned multimodal LLMs for pairwise visual comparison.
- Trained LayoutLM, Donut (Swin+BART), and TrOCR on 100K+ cheques; improved text extraction by 8% and fraud flagging by 4% while reducing inference latency through model optimization.

Machine Learning Engineer | 16 Bit

Oct 2023 - Mar 2024

- Led temporal X-ray image-to-image translation between baseline and follow-up scans (12-96 months) to study knee osteoarthritis progression; scaled GAN and diffusion experiments with PyTorch Lightning/LoRA on 70K+ images, achieving ~10x faster training and ~19% higher image realism.

SELECTED PROJECTS

Echo Charlie: multimodal voice restoration system; Boson AI Hackathon winner (2025).

2D Sketch to 3D Stylization: NeRF-based sketch stylization through image style transfer and CLIP-guided text prompts.

MobileGen: optimized DDPM for mobile inference using quantization-aware training, distillation, and pruning; 20% faster with 14% fewer GFLOPs.